PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
planets = {
    "Mercury": {"order": 1, "type": "terrestrial", "moons": 0,  "diameter_km": 4879},
    "Venus":   {"order": 2, "type": "terrestrial", "moons": 0,  "diameter_km": 12104},
    "Earth":   {"order": 3, "type": "terrestrial", "moons": 1,  "diameter_km": 12742},
    "Mars":    {"order": 4, "type": "terrestrial", "moons": 2,  "diameter_km": 6779},
    "Jupiter": {"order": 5, "type": "gas giant",   "moons": 95, "diameter_km": 139820},
    "Saturn":  {"order": 6, "type": "gas giant",   "moons": 146, "diameter_km": 116460},
    "Uranus":  {"order": 7, "type": "ice giant",   "moons": 28, "diameter_km": 50724},
    "Neptune": {"order": 8, "type": "ice giant",   "moons": 16, "diameter_km": 49244},
}
has_rings = {"Jupiter", "Saturn", "Uranus", "Neptune"}
def planet_at(n):          return next(p for p, d in planets.items() if d["order"] == n)
def is_inner(p):           return planets[p]["order"] <= 4
def is_outer(p):           return not is_inner(p)
def is_larger(a, b):       return planets[a]["diameter_km"] > planets[b]["diameter_km"]
def of_type(t):            return [p for p, d in planets.items() if d["type"] == t]
def next_planet(p):
    n = planets[p]["order"] + 1
    return planet_at(n) if n <= 8 else None
print("Third planet from the Sun      :", planet_at(3))
print("Is Mars an inner planet?       ", is_inner("Mars"))
print("Is Saturn an outer planet?     ", is_outer("Saturn"))
print("Is Jupiter larger than Earth?  ", is_larger("Jupiter", "Earth"))
print("Planet after Earth             :", next_planet("Earth"))
print("Terrestrial planets            :", of_type("terrestrial"))
print("Gas giants                     :", of_type("gas giant"))
print("Planets with no moons          :", [p for p, d in planets.items() if d["moons"] == 0])
print("Planets with rings             :", sorted(has_rings))
print("Largest planet                 :", max(planets, key=lambda p: planets[p]["diameter_km"]))
```

OUTPUT:

```
>>>
    ========== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ==========
    Third planet from the Sun      : Earth
    Is Mars an inner planet?        True
    Is Saturn an outer planet?      True
    Is Jupiter larger than Earth?   True
    Planet after Earth             : Mars
    Terrestrial planets            : ['Mercury', 'Venus', 'Earth', 'Mars']
    Gas giants                     : ['Jupiter', 'Saturn']
    Planets with no moons          : ['Mercury', 'Venus']
    Planets with rings             : ['Jupiter', 'Neptune', 'Saturn', 'Uranus']
    Largest planet                 : Jupiter
>>>
```